

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3281

3 Credits: 3

Program Core Course

Source-grounded

Fundamentals of Cryptography and

□ To understand basics and fundamentals of Information Security and Cryptography. □ To understand the need of security

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3281 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3281.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Cyber Forensics And Information Security
Course code	3281
Course title	Fundamentals of Cryptography and
Semester	3 Credits: 3
Course category	Program Core Course
Periods per week	3 (L:3 T:0 P:0) Periods per semester: 45
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

26 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To understand basics and fundamentals of Information Security and Cryptography. To understand the need of security
- To identify various malicious softwares and understand the intrusion detection
- To identify and mitigate DoS attacks and the use of firewalls.

Course outcomes

CO	Official outcome	Study level
CO1	11 Understanding Information security and Cryptography	See official syllabus
CO2	Illustrate the various methods of user authentication 11 Understanding Classify various malicious software and explain	See official syllabus
CO3	9 Understanding Intrusion detection systems	See official syllabus
CO4	Understand Denial of service and Firewalls 12 Understanding	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3 2
CO3	CO3 3 2
CO4	CO4 3 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Cryptography	8	As specified
Module 2	Illustrate the various methods of user authentication	9	As specified
Module 3	systems	5	As specified
Module 4	Understand Denial of service and Firewalls	4	As specified

MODULE 1

Cryptography

This module is presented from the official syllabus scope for Fundamentals of Cryptography and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Cryptography
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

▼ Explain the history of Information Security 1 Understanding Explain the components of an Information

Explain the history of Information Security 1 Understanding Explain the components of an Information is an official topic in Module 1 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the history of Information Security 1 Understanding Explain the components of an Information using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the history of information security 1 understanding explain the components of an information should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the history of Information Security 1 Understanding Explain the components of an Information means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the history of Information Security 1 Understanding Explain the components of an Information definition/meaning , steps, application Technical terms English-

▼ 1 Understanding system. Illustrate the need for security, Distinguish

1 Understanding system. Illustrate the need for security, Distinguish is an official topic in Module 1 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding system. Illustrate the need for security, Distinguish using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding system. illustrate the need for security, distinguish should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Malayalam support: Module 1- 1 Understanding system. Illustrate the need for security, Distinguish between definition/meaning and application. Explain the steps, application of security. Technical terms English- Malayalam.

▼ 1 Understanding Threats and Attacks, Threats and Assets Explain classical encryption techniques and

1 Understanding Threats and Attacks, Threats and Assets Explain classical encryption techniques and is an official topic in Module 1 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Threats and Attacks, Threats and Assets Explain classical encryption techniques and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding threats and attacks, threats and assets explain classical encryption techniques and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Threats and Attacks, Threats and Assets Explain classical encryption techniques and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding Threats and Attacks, Threats and Assets Explain classical encryption techniques and definition/meaning. , steps, application . Technical terms English- .

▼ 2 Understanding Symmetric cipher modes Understanding

2 Understanding Symmetric cipher modes Understanding is an official topic in Module 1 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Symmetric cipher modes Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding symmetric cipher modes understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Symmetric cipher modes Understanding means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding Symmetric cipher modes Understanding definition/meaning . , steps, application . Technical terms English- .

▼ Explain various substitution techniques 2

Explain various substitution techniques 2 is an official topic in Module 1 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain various substitution techniques 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain various substitution techniques 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain various substitution techniques 2 means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain various substitution techniques 2
 പരമ്പരാഗതമായും ആധുനികമായും definition/meaning ഉൾപ്പെടെയുള്ളവ. പരമ്പരാഗതമായും ആധുനികമായും
 പരമ്പരാഗതമായും, steps, application ഉൾപ്പെടെയുള്ളവ ഉൾപ്പെടെയുള്ളവ. Technical terms
 English-ൽ ഉൾപ്പെടെയുള്ളവ ഉൾപ്പെടെയുള്ളവ.

▼ Explain various transposition techniques 2 Understanding Demonstrate the use of Steganography in Understanding

Explain various transposition techniques 2 Understanding Demonstrate the use of Steganography in Understanding is an official topic in Module 1 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain various transposition techniques 2 Understanding Demonstrate the use of Steganography in Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain various transposition techniques 2 understanding demonstrate the use of steganography in understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain various transposition techniques 2 Understanding Demonstrate the use of Steganography in Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain various transposition techniques 2 Understanding Demonstrate the use of Steganography in Understanding definition/meaning , steps, application Technical terms English-

▼ 2 Cryptography Explain simplified model of symmetric Understanding

2 Cryptography Explain simplified model of symmetric Understanding is an official topic in Module 1 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Cryptography Explain simplified model of symmetric Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 cryptography explain simplified model of symmetric understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Cryptography Explain simplified model of symmetric Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Cryptography Explain simplified model of symmetric Understanding definition/meaning . , steps, application . Technical terms English- .

▼ 1 encryption (basic idea only) Computer Security : History of Information Security-Components of Information system- Definition - Triad - Confidentiality , Integrity, Availability - Need of Security concepts - relationships - Threats , Attack, Assets Cryptography : Classical Encryption Techniques- Symmetric cipher modes- Substitution techniques - Transposition techniques - Steganography - Symmetric Encryption

1 encryption (basic idea only) Computer Security : History of Information Security-Components of Information system- Definition -Triad - Confidentiality , Integrity, Availability - Need of Security concepts - relationships - Threats , Attack, Assets Cryptography : Classical Encryption Techniques- Symmetric cipher modes- Substitution techniques - Transposition techniques - Steganography - Symmetric Encryption is an official topic in Module 1 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 encryption (basic idea only) Computer Security : History of Information Security-Components of Information system- Definition -Triad - Confidentiality , Integrity, Availability - Need of Security concepts - relationships - Threats , Attack, Assets Cryptography : Classical Encryption Techniques- Symmetric cipher modes- Substitution techniques - Transposition techniques - Steganography - Symmetric Encryption using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 encryption (basic idea only) computer security : history of information security-components of information system- definition -triad - confidentiality , integrity, availability - need of security concepts - relationships - threats , attack, assets cryptography : classical encryption techniques- symmetric cipher modes- substitution techniques - transposition techniques - steganography - symmetric encryption should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 encryption (basic idea only) Computer Security : History of Information Security-Components of Information system- Definition - Triad - Confidentiality , Integrity, Availability - Need of Security concepts - relationships - Threats , Attack, Assets Cryptography : Classical Encryption Techniques- Symmetric cipher modes- Substitution techniques - Transposition techniques - Steganography - Symmetric Encryption means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 encryption (basic idea only) Computer Security : History of Information Security-Components of Information system- Definition -Triad - Confidentiality , Integrity, Availability - Need of Security concepts - relationships - Threats , Attack, Assets Cryptography : Classical Encryption Techniques- Symmetric cipher modes- Substitution techniques - Transposition techniques - Steganography - Symmetric Encryption definition/meaning . steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Illustrate the various methods of user authentication

This module is presented from the official syllabus scope for Fundamentals of Cryptography and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the various methods of user authentication
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding Authentication and the means of authentication

1 Understanding Authentication and the means of authentication is an official topic in Module 2 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Authentication and the means of authentication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding authentication and the means of authentication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding Authentication and the means of authentication means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding Authentication and the means of authentication ഏതു ഏതു definition/meaning ഉൾപ്പെടെ. ഏതു ഏതു steps, application ഉൾപ്പെടെ. Technical terms English-ൽ ഉൾപ്പെടെ.

▼ Explain Password based Authentication 1 Understanding Explain various Password attack strategies and

Explain Password based Authentication 1 Understanding Explain various Password attack strategies and is an official topic in Module 2 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Password based Authentication 1 Understanding Explain various Password attack strategies and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain password based authentication 1 understanding explain various password attack strategies and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Password based Authentication 1 Understanding Explain various Password attack strategies and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain Password based Authentication 1 Understanding Explain various Password attack strategies and definition/meaning. Steps, application. Technical terms English-.

▼ 2 Understanding countermeasures

2 Understanding countermeasures is an official topic in Module 2 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding countermeasures using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding countermeasures should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding countermeasures means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Understanding countermeasures
 പരമ്പരാഗതമായി നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക
 നിർവ്വചനം, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms
 English-ൽ നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

▼ Describe the use of hashed passwords 1 Understanding Explain various password cracking approaches

Describe the use of hashed passwords 1 Understanding Explain various password cracking approaches is an official topic in Module 2 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the use of hashed passwords 1 Understanding Explain various password cracking approaches using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the use of hashed passwords 1 understanding explain various password cracking approaches should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the use of hashed passwords 1 Understanding Explain various password cracking approaches means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Describe the use of hashed passwords 1 Understanding Explain various password cracking approaches definition/meaning . steps, application . Technical terms English- .

▼ 2 Understanding and also about user password choices Explain the basic concepts of password File

2 Understanding and also about user password choices Explain the basic concepts of password File is an official topic in Module 2 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding and also about user password choices Explain the basic concepts of password File using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding and also about user password choices explain the basic concepts of password file should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding and also about user password choices Explain the basic concepts of password File means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Understanding and also about user password choices Explain the basic concepts of password File definition/meaning steps, application Technical terms English-

▼ 1 Understanding access control

1 Understanding access control is an official topic in Module 2 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding access control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding access control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding access control means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-1 Understanding access control
 പരീക്ഷയ്ക്കായി definition/meaning നൽകുക. പരീക്ഷയിൽ പരീക്ഷിക്കുക
 പരീക്ഷിക്കുക, steps, application നൽകുക പരീക്ഷിക്കുക പരീക്ഷിക്കുക. Technical terms
 English-ൽ പരീക്ഷിക്കുക പരീക്ഷിക്കുക.

▼ Outline various password selection strategies 1 Understanding Outline various Authentication Methods such as Token based authentication , Biometric

Outline various password selection strategies 1 Understanding Outline various Authentication Methods such as Token based authentication , Biometric is an official topic in Module 2 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline various password selection strategies 1 Understanding Outline various Authentication Methods such as Token based authentication , Biometric using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline various password selection strategies 1 understanding outline various authentication methods such as token based authentication , biometric should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline various password selection strategies 1 Understanding Outline various Authentication Methods such as Token based authentication , Biometric means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Outline various password selection strategies 1 Understanding Outline various Authentication Methods such as Token based authentication , Biometric definition/meaning . , steps, application . Technical terms English- .

▼ 1 Understanding Authentication , Remote User Authentication.(Basic idea only) Relate Access control in computer security,

1 Understanding Authentication , Remote User Authentication.(Basic idea only) Relate Access control in computer security, is an official topic in Module 2 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Authentication , Remote User Authentication.(Basic idea only) Relate Access control in computer security, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding authentication , remote user authentication.(basic idea only) relate access control in computer security, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Authentication , Remote User Authentication. (Basic idea only) Relate Access control in computer security, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding Authentication , Remote User Authentication.(Basic idea only) Relate Access control in computer security, definition/meaning .
 , steps, application .
 Technical terms English- .

▼ Explain Access Control Principles and 2 Understanding various access control policies User Authentication: Means of authentication - Password based Authentication - Password attack strategies and countermeasures - hashed passwords - password cracking - user password choices - password File access control - password selection. Authentication Methods: Token based authentication -Biometric Authentication - Remote User Authentication - security issues- Access control: Principles - Relationship among other security functions - access control policies Classify various malicious software and explain Intrusion detection

Explain Access Control Principles and 2 Understanding various access control policies User Authentication: Means of authentication - Password based Authentication - Password attack strategies and countermeasures - hashed passwords - password cracking - user password choices - password File access control - password selection. Authentication Methods: Token based authentication -Biometric Authentication - Remote User Authentication - security issues- Access control: Principles - Relationship among other security functions - access control policies Classify various malicious software and explain Intrusion detection is an official topic in Module 2 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Access Control Principles and 2 Understanding various access control policies User Authentication: Means of authentication

- Password based Authentication - Password attack strategies and countermeasures - hashed passwords - password cracking - user password choices - password File access control - password selection. Authentication Methods: Token based authentication -Biometric Authentication - Remote User Authentication - security issues- Access control: Principles - Relationship among other security functions - access control policies Classify various malicious software and explain Intrusion detection using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain access control principles and 2 understanding various access control policies user authentication: means of authentication - password based authentication - password attack strategies and countermeasures - hashed passwords - password cracking - user password choices - password file access control - password selection. authentication methods: token based authentication -biometric authentication - remote user authentication - security issues- access control: principles - relationship among other security functions - access control policies classify various malicious software and explain intrusion detection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Access Control Principles and 2 Understanding various access control policies User Authentication: Means of authentication - Password based Authentication - Password attack strategies and countermeasures - hashed passwords - password cracking - user password choices - password File access control - password selection. Authentication Methods: Token based authentication -Biometric Authentication - Remote User Authentication - security issues- Access

Aspect	What to prepare
	control: Principles – Relationship among other security functions - access control policies Classify various malicious software and explain Intrusion detection means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain Access Control Principles and 2 Understanding various access control policies User Authentication: Means of authentication - Password based Authentication - Password attack strategies and countermeasures - hashed passwords - password cracking - user password choices - password File access control - password selection. Authentication Methods: Token based authentication -Biometric Authentication - Remote User Authentication - security issues- Access control: Principles – Relationship among other security functions - access control policies Classify various malicious software and explain Intrusion detection definition/meaning. steps, application. Technical terms English- Malayalam.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3**systems**

This module is presented from the official syllabus scope for Fundamentals of Cryptography and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: systems
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ Explain Viruses , The nature of viruses, Virus 2 structure, . Classify Viruses, Antivirus approaches &

Explain Viruses , The nature of viruses, Virus 2 structure, . Classify Viruses, Antivirus approaches & is an official topic in Module 3 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Viruses , The nature of viruses, Virus 2 structure, . Classify Viruses, Antivirus approaches & using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain viruses , the nature of viruses, virus 2 structure, . classify viruses, antivirus approaches & should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Viruses , The nature of viruses, Virus 2 structure, . Classify Viruses, Antivirus approaches & means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain Viruses , The nature of viruses, Virus 2 structure, . Classify Viruses, Antivirus approaches & definition/meaning . steps, application . Technical terms English- .

▼ 1 Understanding Antivirus techniques . Explain Worms- Outline Worm propagation

1 Understanding Antivirus techniques . Explain Worms- Outline Worm propagation is an official topic in Module 3 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Antivirus techniques . Explain Worms- Outline Worm propagation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding antivirus techniques . explain worms-outline worm propagation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Malayalam support: Module 3- 1 Understanding Antivirus techniques . Explain Worms- Outline Worm propagation definition/meaning . steps, application . Technical terms English- .

▼ model - Explain various countermeasures for 2 Understanding worm propagation Outline various classes of intruders and the

model - Explain various countermeasures for 2 Understanding worm propagation Outline various classes of intruders and the is an official topic in Module 3 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify model - Explain various countermeasures for 2 Understanding worm propagation Outline various classes of intruders and the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, model - explain various countermeasures for 2 understanding worm propagation outline various classes of intruders and the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What model - Explain various countermeasures for 2 Understanding worm propagation Outline various classes of intruders and the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-റൂട്ട് കീ model - Explain various countermeasures for 2 Understanding worm propagation Outline various classes of intruders and the ഓരോന്നിന്റെയും definition/meaning ഉൾപ്പെടെ. ഓരോന്നിന്റെയും steps, application ഉൾപ്പെടെ. Technical terms English-ൽ ഉൾപ്പെടെ ഉൾപ്പെടെ.

▼ 2 Understanding intruder behavior patterns. Explain Intrusion Detection System,

2 Understanding intruder behavior patterns. Explain Intrusion Detection System, is an official topic in Module 3 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding intruder behavior patterns. Explain Intrusion Detection System, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding intruder behavior patterns. explain intrusion detection system, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding intruder behavior patterns. Explain Intrusion Detection System, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding intruder behavior patterns. Explain Intrusion Detection System, definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 2 Understanding classification and the requirements of IDS. Malicious Software & Intrusion Detection Malicious Software: Different types - Viruses - nature - Virus structure - Classification - Antivirus approaches - Antivirus techniques - Worms - Propagation model - Worm Countermeasures - Intrusion and Detection: Classes of intruders - Intruder behavior patterns - classification - requirements of IDS

2 Understanding classification and the requirements of IDS. Malicious Software & Intrusion Detection Malicious Software: Different types - Viruses - nature - Virus structure - Classification - Antivirus approaches - Antivirus techniques - Worms - Propagation model - Worm Countermeasures - Intrusion and Detection: Classes of intruders - Intruder behavior patterns - classification - requirements of IDS is an official topic in Module 3 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding classification and the requirements of IDS. Malicious Software & Intrusion Detection Malicious Software: Different types - Viruses - nature - Virus structure - Classification - Antivirus approaches - Antivirus techniques - Worms - Propagation model - Worm Countermeasures - Intrusion and Detection: Classes of intruders - Intruder behavior patterns - classification - requirements of IDS using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding classification and the requirements of ids. malicious software & intrusion detection malicious software: different types - viruses - nature - virus structure - classification - antivirus approaches - antivirus techniques - worms - propagation model - worm countermeasures - intrusion and detection: classes of intruders - intruder behavior patterns - classification - requirements of ids should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding classification and the requirements of IDS. Malicious Software & Intrusion Detection Malicious Software: Different types - Viruses - nature - Virus structure - Classification - Antivirus approaches - Antivirus techniques - Worms - Propagation model - Worm Countermeasures - Intrusion and Detection: Classes of intruders - Intruder behavior patterns - classification - requirements of IDS means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding classification and the requirements of IDS. Malicious Software & Intrusion Detection Malicious Software: Different types - Viruses - nature - Virus structure - Classification - Antivirus approaches - Antivirus techniques - Worms - Propagation model - Worm Countermeasures - Intrusion and Detection: Classes of intruders - Intruder behavior patterns - classification - requirements of IDS

മലയാള സപ്പോർട്ട് മോഡ്യൂൾ 3- 2 മനസ്സിലാക്കൽ വർഗ്ഗീകരണം ആവശ്യകതകൾ. ദുഷകരമായ സോഫ്റ്റ്‌വെയർ & ഇൻട്രൂഷൻ ഡിറ്റക്ഷൻ ദുഷകരമായ സോഫ്റ്റ്‌വെയർ: വ്യത്യസ്ത തരങ്ങൾ - വൈറസ് - സ്വഭാവം - വൈറസ് ഘടന - വർഗ്ഗീകരണം - ആന്റിവൈറസ് സമീപനങ്ങൾ - ആന്റിവൈറസ് രീതികൾ - വേർമ - പ്രചാരണ മോഡൽ - വേർമ കൗണ്ടർമേച്ചറുകൾ - ഇൻട്രൂഷൻ ആന്റ് ഡിറ്റക്ഷൻ: ഇൻട്രൂഡർ ക്ലാസ്സുകൾ - ഇൻട്രൂഡർ പെരുമാറ്റ പാറ്റേണുകൾ - വർഗ്ഗീകരണം - വർഗ്ഗീകരണം ആവശ്യകതകൾ. Technical terms English-ൽ മലയാളത്തിൽ പരിഭാഷിപ്പിക്കുക.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Understand Denial of service and Firewalls

This module is presented from the official syllabus scope for Fundamentals of Cryptography and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand Denial of service and Firewalls
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ of DoS on Network bandwidth, System 2 Understanding resources and Application resources

of DoS on Network bandwidth, System 2 Understanding resources and Application resources is an official topic in Module 4 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify of DoS on Network bandwidth, System 2 Understanding resources and Application resources using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, of dos on network bandwidth, system 2 understanding resources and application resources should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What of DoS on Network bandwidth, System 2 Understanding resources and Application resources means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- of DoS on Network bandwidth, System 2 Understanding resources and Application resources definition/meaning . . . , steps, application Technical terms English-

▼ Interpret the Classic Denial of Service Attacks 2 Understanding Explain the need of firewall and summarize

Interpret the Classic Denial of Service Attacks 2 Understanding Explain the need of firewall and summarize is an official topic in Module 4 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Interpret the Classic Denial of Service Attacks 2 Understanding Explain the need of firewall and summarize using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interpret the classic denial of service attacks 2 understanding explain the need of firewall and summarize should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Interpret the Classic Denial of Service Attacks 2 Understanding Explain the need of firewall and summarize means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Interpret the Classic Denial of Service Attacks 2 Understanding Explain the need of firewall and summarize definition/meaning. , steps, application . Technical terms English- .

▼ 2 Understanding various characteristics of a Firewall

2 Understanding various characteristics of a Firewall is an official topic in Module 4 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding various characteristics of a Firewall using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding various characteristics of a firewall should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding various characteristics of a Firewall means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Understanding various characteristics of a Firewall
 definition/meaning
 steps, application
 Technical terms English-

▼ Classify firewalls. 2 Understanding Denial of Service and Firewall

Denial of Service: Definition - Effect of DoS on Network bandwidth and System resources -Classic Denial of Service Attacks -: Need -

Characteristics - Types of firewalls . Text / Reference T/R Book

Title/Author Computer Security- Principles and Practice - Author:

William Stallings & Lawrie T1 Brown Publisher: Pearson Prentice Hall

2010 Cryptography And Network Security Principles And Practice

Fourth Edition, T2 William Stallings, Pearson Education R1 W. Mao,

“Modern Cryptography – Theory and Practice”, Pearson Education. R2

Charles P. Pfleeger, Shari Lawrence Pfleeger – Security in computing -

Prentice Hall of India. Online Resources SI No Website Link 1

1 https://www.tutorialspoint.com/cryptography/pdf/cryptography_quick_guide.pdf

2 https://www.w3schools.com/cybersecurity/cybersecurity_firewalls.php

3 <https://www.omnisecu.com/security/index.php>

Classify firewalls. 2 Understanding Denial of Service and Firewall Denial of Service: Definition - Effect of DoS on Network bandwidth and System resources -Classic Denial of Service Attacks -: Need - Characteristics - Types of firewalls . Text / Reference T/R Book Title/Author Computer Security- Principles and Practice - Author: William Stallings & Lawrie T1 Brown Publisher: Pearson Prentice Hall 2010 Cryptography And Network Security Principles And Practice Fourth Edition, T2 William Stallings, Pearson Education R1 W. Mao, “Modern Cryptography – Theory and Practice”, Pearson Education. R2 Charles P. Pfleeger, Shari Lawrence Pfleeger – Security in computing –Prentice Hall of India. Online Resources SI No Website Link 1

https://www.tutorialspoint.com/cryptography/pdf/cryptography_quick_guide.pdf

2 https://www.w3schools.com/cybersecurity/cybersecurity_firewalls.php 3

<https://www.omnisecu.com/security/index.php> is an official topic in Module

4 of Fundamentals of Cryptography and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify firewalls. 2 Understanding Denial of Service and Firewall Denial of Service: Definition - Effect of DoS on Network bandwidth and System resources -Classic Denial of Service Attacks -: Need - Characteristics - Types of firewalls . Text / Reference T/R Book Title/Author Computer Security- Principles and Practice - Author: William Stallings & Lawrie T1 Brown Publisher: Pearson Prentice Hall 2010 Cryptography And Network Security Principles And Practice Fourth Edition, T2 William Stallings, Pearson Education R1 W. Mao, “Modern Cryptography – Theory and Practice”, Pearson Education. R2 Charles P. Pfleeger, Shari Lawrence Pfleeger – Security in computing –Prentice Hall of India. Online Resources SI No Website Link 1 https://www.tutorialspoint.com/cryptography/pdf/cryptography_quick_guide.pdf 2 https://www.w3schools.com/cybersecurity/cybersecurity_firewalls.php 3 <https://www.omnisecu.com/security/index.php> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify firewalls. 2 understanding denial of service and firewall denial of service: definition - effect of dos on network bandwidth and system resources -classic denial of service attacks -: need - characteristics - types of firewalls . text / reference t/r book title/author computer security- principles and practice - author: william stallings & lawrie t1 brown publisher: pearson prentice hall 2010 cryptography and network security principles and

practice fourth edition, t2 william stallings, pearson education r1 w. mao, “modern cryptography – theory and practice”, pearson education. r2 charles p. pfleeger, shari lawrence pfleeger – security in computing –prentice hall of india. online resources sl no website link 1

https://www.tutorialspoint.com/cryptography/pdf/cryptography_quick_guide.pdf 2

https://www.w3schools.com/cybersecurity/cybersecurity_firewalls.php 3

<https://www.omnisecu.com/security/index.php> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classify firewalls. 2 Understanding Denial of Service and Firewall Denial of Service: Definition - Effect of DoS on Network bandwidth and System resources -Classic Denial of Service Attacks -: Need - Characteristics - Types of firewalls . Text / Reference T/R Book Title/Author Computer Security- Principles and Practice - Author: William Stallings & Lawrie T1 Brown Publisher: Pearson Prentice Hall 2010 Cryptography And Network Security Principles And Practice Fourth Edition, T2 William Stallings, Pearson Education R1 W. Mao, “Modern Cryptography – Theory and Practice”, Pearson Education. R2 Charles P. Pfleeger, Shari Lawrence Pfleeger – Security in computing –Prentice Hall of India. Online Resources SI No Website Link 1 https://www.tutorialspoint.com/cryptography/pdf/cryptography_quick_guide.p 2 https://www.w3schools.com/cybersecurity/cybersecurity_firewalls.php 3 https://www.omnisecu.com/security/index.php means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic

with a related but different term.

Malayalam support: Module 4-☐☐ Classify firewalls. 2 Understanding Denial of Service and Firewall Denial of Service: Definition - Effect of DoS on Network bandwidth and System resources -Classic Denial of Service Attacks -: Need - Characteristics - Types of firewalls . Text / Reference T/R Book Title/Author Computer Security- Principles and Practice - Author: William Stallings & Lawrie T1 Brown Publisher: Pearson Prentice Hall 2010 Cryptography And Network Security Principles And Practice Fourth Edition, T2 William Stallings, Pearson Education R1 W. Mao, “Modern Cryptography – Theory and Practice”, Pearson Education. R2 Charles P. Pfleeger, Shari Lawrence Pfleeger – Security in computing –Prentice Hall of India. Online Resources SI No Website Link 1 https://www.tutorialspoint.com/cryptography/pdf/cryptography_quick_guide.pdf 2 https://www.w3schools.com/cybersecurity/cybersecurity_firewalls.php 3 <https://www.omnisecl.com/security/index.php> ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain Explain the history of Information Security 1 Understanding Explain the components of an Information. (3 marks)**

► **Define or explain 1 Understanding system. Illustrate the need for security, Distinguish. (3 marks)**

► **Define or explain 1 Understanding Threats and Attacks, Threats and Assets Explain classical encryption techniques and. (3 marks)**

► **Define or explain 2 Understanding Symmetric cipher modes Understanding. (3 marks)**

Module 2

► **Define or explain 1 Understanding Authentication and the means of authentication. (3 marks)**

► **Define or explain Explain Password based Authentication 1 Understanding Explain various Password attack strategies and. (3 marks)**

► **Define or explain 2 Understanding countermeasures. (3 marks)**

- ▶ **Define or explain Describe the use of hashed passwords 1 Understanding Explain various password cracking approaches. (3 marks)**

Module 3

- ▶ **Define or explain Explain Viruses , The nature of viruses, Virus 2 structure, . Classify Viruses, Antivirus approaches &. (3 marks)**

- ▶ **Define or explain 1 Understanding Antivirus techniques . Explain Worms- Outline Worm propagation. (3 marks)**

- ▶ **Define or explain model - Explain various countermeasures for 2 Understanding worm propagation Outline various classes of intruders and the. (3 marks)**

- ▶ **Define or explain 2 Understanding intruder behavior patterns. Explain Intrusion Detection System,. (3 marks)**

Module 4

- ▶ **Define or explain of DoS on Network bandwidth, System 2 Understanding resources and Application resources. (3 marks)**

- ▶ **Define or explain Interpret the Classic Denial of Service Attacks 2 Understanding Explain the need of firewall and summarize. (3 marks)**

- ▶ **Define or explain 2 Understanding various characteristics of a Firewall. (3 marks)**

- ▶ **Define or explain Classify firewalls. 2 Understanding Denial of Service and Firewall Denial of Service: Definition - Effect of DoS on Network bandwidth and System resources -Classic Denial of Service Attacks -:**

Need - Characteristics - Types of firewalls . Text / Reference T/R Book
Title/Author Computer Security- Principles and Practice - Author:
William Stallings & Lawrie T1 Brown Publisher: Pearson Prentice Hall
2010 Cryptography And Network Security Principles And Practice Fourth
Edition, T2 William Stallings, Pearson Education R1 W. Mao, “Modern
Cryptography - Theory and Practice”, Pearson Education. R2 Charles P.
Pfleeger, Shari Lawrence Pfleeger - Security in computing -Prentice Hall
of India. Online Resources SI No Website Link 1
1 https://www.tutorialspoint.com/cryptography/pdf/cryptography_quick_guide.pdf
2 https://www.w3schools.com/cybersecurity/cybersecurity_firewalls.php
3 <https://www.omnisecu.com/security/index.php>. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain the history of Information Security 1 Understanding Explain the components of an Information.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **1 Understanding Authentication and the means of authentication.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain Viruses , The nature of viruses, Virus 2 structure, . Classify Viruses, Antivirus approaches &.**

Module 4

1-mark definition: State the meaning of **of DoS on Network bandwidth, System 2 Understanding resources and Application resources.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- SI No Website Link
- https://www.tutorialspoint.com/cryptography/pdf/cryptography_quick_guide.pdf
- https://www.w3schools.com/cybersecurity/cybersecurity_firewalls.php
<https://www.omnisecu.com/security/index.php>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 3281. Verify institutional updates against the current official curriculum.